

LaserCortex

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Chapter 1

EML Registry: Tamari Lattice Contraction

1.1 Core Types

Definition 1 (EML Tree). The type of binary trees representing configurations in the Tamari lattice. A tree is either a **Leaf** (empty configuration) or a **Node** $t_1\ t_2$ (composition of two sub-configurations).

Definition 2 (Tree Size). The number of internal nodes in a tree, used as the lattice index n in T_n .

Definition 3 (Single-Step Contraction). The primitive non-associative rewrite rule: $(a \bullet b) \bullet c \rightarrow a \bullet (b \bullet c)$. This is the coupling signature for choice resolution, lifted through subtrees.

Definition 4 (Multi-Step Contraction). The reflexive-transitive closure of `contracts_one`. Represents an audit trail of choice resolutions with monotonic provenance.

Definition 5 (Right-Comb Normal Form). The minimum element of the Tamari lattice — the equilibrium attractor. Defined as `rightComb(0) = Leaf` and `rightComb(n+1) = Node(Leaf, rightComb(n))`.

1.2 Lifting Lemmas

Lemma 6 (Contraction Preserved Under Left Node). *If $t \rightarrow t'$ then $\text{Node}(t, r) \rightarrow \text{Node}(t', r)$. Choice resolution in the left subtree propagates to the whole composition.*

Lemma 7 (Contraction Preserved Under Right Node). *If $r \rightarrow r'$ then $\text{Node}(l, r) \rightarrow \text{Node}(l, r')$. Choice resolution in the right subtree propagates to the whole composition.*

Lemma 8 (Transitivity of Contraction). *If $s \rightarrow t$ and $t \rightarrow u$ then $s \rightarrow u$. Audit trails concatenate monotonically.*

1.3 Size Invariants

Lemma 9 (Size is Preserved by Single-Step Contraction). *Rotation rewrites the tree depth but does not change the number of internal nodes.*

Lemma 10 (Size is Preserved by Multi-Step Contraction). *Contraction paths preserve tree size, establishing that the Tamari lattice T_n is graded by node count n .*

1.4 Main Contraction Theorem

Lemma 11 (Node of Right-Combs Contracts to Right-Comb). *For any $a, b : \mathbb{N}$,*

$$\text{Node}(\text{rightComb}(a), \text{rightComb}(b)) \longrightarrow \text{rightComb}(1 + a + b).$$

The equilibrium attractor is closed under non-associative composition.

Theorem 12 (Every Tree Contracts to Right-Comb). *For any tree $t : \text{EMLTree}$,*

$$t \longrightarrow \text{rightComb}(t.\text{size}).$$

Every configuration has a valid evolution path to the equilibrium attractor indexed by its size.

1.5 Router and Registry

Definition 13 (Router Index). A bounded natural number $\text{Fin}(n)$ serving as a neural binding address.

Definition 14 (Type Registry). An injective mapping from router indices to EML trees. This is the cortex-registry interface where neural computation meets formal grammar.

1.6 Witness-Skeptic Game

Definition 15 (Cortex Certificate). A proof-carrying audit trail $\langle \text{source}, \text{target}, \text{proof} \rangle$ showing that a tree reaches its equilibrium.

Definition 16 (Certify). Issues a `CortexCertificate` for any tree t , certifying that $t \rightarrow \text{rightComb}(t.\text{size})$. This is the quench witness in the Witness-Skeptic game.

Chapter 2

Logic Types: Pluralistic Reasoning Framework

2.1 Logic Type Hierarchy

Definition 17 (13 Logic Types). An inductive type enumerating 13 reasoning paradigms. Each handles specific classes of paradoxes as boundary conditions rather than errors.

Definition 18 (Logic Classification). Classification of logic types as Substructural, Extension, Alternative, Generalization, or Classical.

Definition 19 (Logic Tree). A type family assigning EML trees to each logic type. Currently uniform; extensible to logic-specific tree types.

Definition 20 (Logic Contraction). A type family of contraction relations indexed by `LogicType`. For Classical logic this is the Tamari contraction from `EMLRegistry`.

Definition 21 (Logic Translation). A structure specifying forward and backward maps between logic types with soundness and completeness proofs.

Definition 22 (Meta-Contraction). Cross-logic contraction allowing reasoning that spans multiple logic types via intra-logic contraction, inter-logic translation, and transitivity.

2.2 Cayley-Dickson Connection

Definition 23 (CD Step). Maps logic types to Cayley-Dickson construction steps.

Definition 24 (Split-Octonion Sectors). Divides logic types into associative and non-associative sectors of the split-octonion algebra.

2.3 Normal Forms and Contraction Theorem

Definition 25 (Logic Normal Form). The normal form for each logic type. For Classical logic this is `rightComb`.

Theorem 26 (Logic Contraction to Normal Form). *For any logic type lt and tree t , `LogicContraction lt t` (`LogicNormalForm lt t`). Currently proven for Classical logic; other cases are placeholders.*

Chapter 3

Liar Paradox: The Hard Test Case

3.1 Canonical Encoding

Definition 27 (Liar Tree). The canonical Liar sentence: a size-3 symmetric tree

$$\text{Node}(\text{Node}(\text{Leaf}, \text{Leaf}), \text{Node}(\text{Leaf}, \text{Leaf}))$$

encoding the self-referential equation $L = \neg L$ via repeated bracketing. Each leaf pair is structurally identical, so neither bracketing dominates — the paradox is that rotation can pass the “hot potato” back and forth indefinitely.

Theorem 28 (Liar Contracts to Right-Comb).

$$\text{liar} \longrightarrow \text{rightComb}(3)$$

The Liar reaches equilibrium under Classical Tamari contraction, as a special case of the general theorem.

3.2 Confluence

Theorem 29 (Classical Confluence of the Liar). *If $\text{liar} \rightarrow t$ then $t \rightarrow \text{rightComb}(3)$. All Classical contraction paths from the Liar converge to the unique minimum of T_3 .*

Theorem 30 (Cross-Logic Confluence of the Liar (Placeholder)). *If $\text{liar} \rightarrow_{lt_1} t_1$ and $\text{liar} \rightarrow_{lt_2} t_2$, there exists a translation between t_1 and t_2 . This is the **open universe** property: different logics produce different normal forms, connected by translations but not necessarily equal. **Status:** Returns True as a placeholder. The real proof requires *logic-specific contraction relations*, which are not yet defined.*

3.3 Emergent Distance and Friction

Definition 31 (Contraction Cost). The pentagonator distance $\Phi(lt, t)$ for resolving tree t under logic type lt . **Status:** Stub returning 0 for all logics. A proper implementation requires a shortest-path search over the contraction graph for each logic type.

Definition 32 (Friction Lagrangian). The action functional $L(lt, t) = \Phi(lt, t)$ evaluated on a single (logic type, tree) pair. **Status:** Currently identical to `contractionCost`; the full infinite-series form with commutator and associator derivatives is not yet formalized.

Definition 33 (Optimal Resolution). The logic type minimizing the Friction Lagrangian for a given tree. **Status:** Returns `Classical` unconditionally. Requires real cost data.

3.4 Missing Work

The following definitions and theorems are stated but not yet implemented. They represent the **real friction** in the framework — the Liar Paradox is defined and its `Classical` resolution is proven, but cross-logic analysis cannot proceed until these pieces are populated.

3.4.1 Sorries in `LogicTypes.lean`

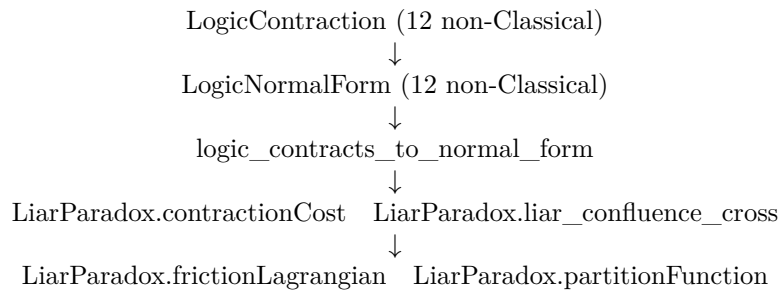
- **LogicContraction** for Fuzzy, ManyValued, Paraconsistent, Temporal, Deontic, Epistemic, Quantum, Intuitionistic, Relevance, Free, Infinitary, Modal — only `Classical` is defined (as the Tamari contraction from `EMLRegistry`).
- **LogicNormalForm** for all non-`Classical` logics — only `rightComb` (`Classical`) is defined.
- **logic_contracts_to_normal_form** for non-`Classical` logics — the theorem is stated but only the `Classical` case is proven.

3.4.2 Stubs in `LiarParadox.lean`

- **contractionCost** — returns 0 for all logics. Needs a bounded shortest-path search over the contraction graph of each logic type, analogous to `decidable_contracts_to` but cost-weighted.
- **frictionLagrangian** — currently the identity on cost. The full form $\sum_k \mu_k [e^{\alpha_k C^{(k)}} - \beta_k \ln(1 + (A^{(k)})^2)]$ with coupling $\sum_k \nu_k (C^{(k)} A^{(k+1)} - A^{(k)} C^{(k+1)})$ is not yet formalized.
- **optimalResolution** — returns `Classical` unconditionally.
- **liar_confluence_cross** — returns `True`; the real proof depends on the missing `LogicContraction` definitions.

3.4.3 Blocking Graph

The dependencies among missing work:



Every missing piece in the chain below `LogicContraction` is blocked by it. The Liar formalization makes this blocking structure explicit: previously the sorries in `LogicTypes.lean` were abstract “to be filled later” — now they are concrete dependencies of the Liar’s cross-logic analysis.